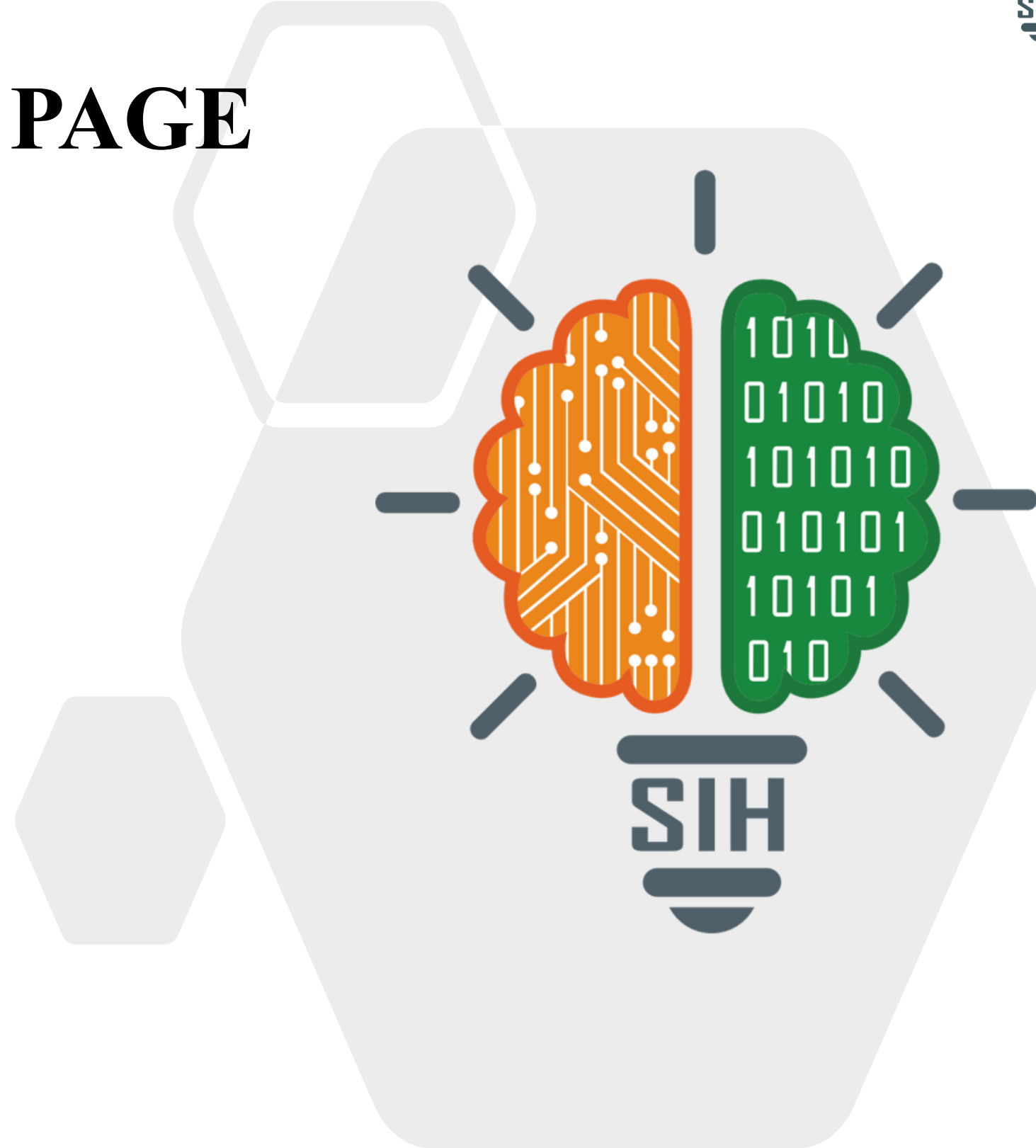


TITLE PAGE



TBD

Problem:

- Disconnected Polar Data Silos: 40+ years of expedition reports, datasets, publications, and media scattered across disparate archives.
- Complex Science Inaccessible to Public: Specialized cryospheric and oceanographic research is difficult for students and citizens to consume.
- Manual & Slow Outreach Creation: Disseminating verified scientific findings across web, social media, and press requires tedious manual effort.

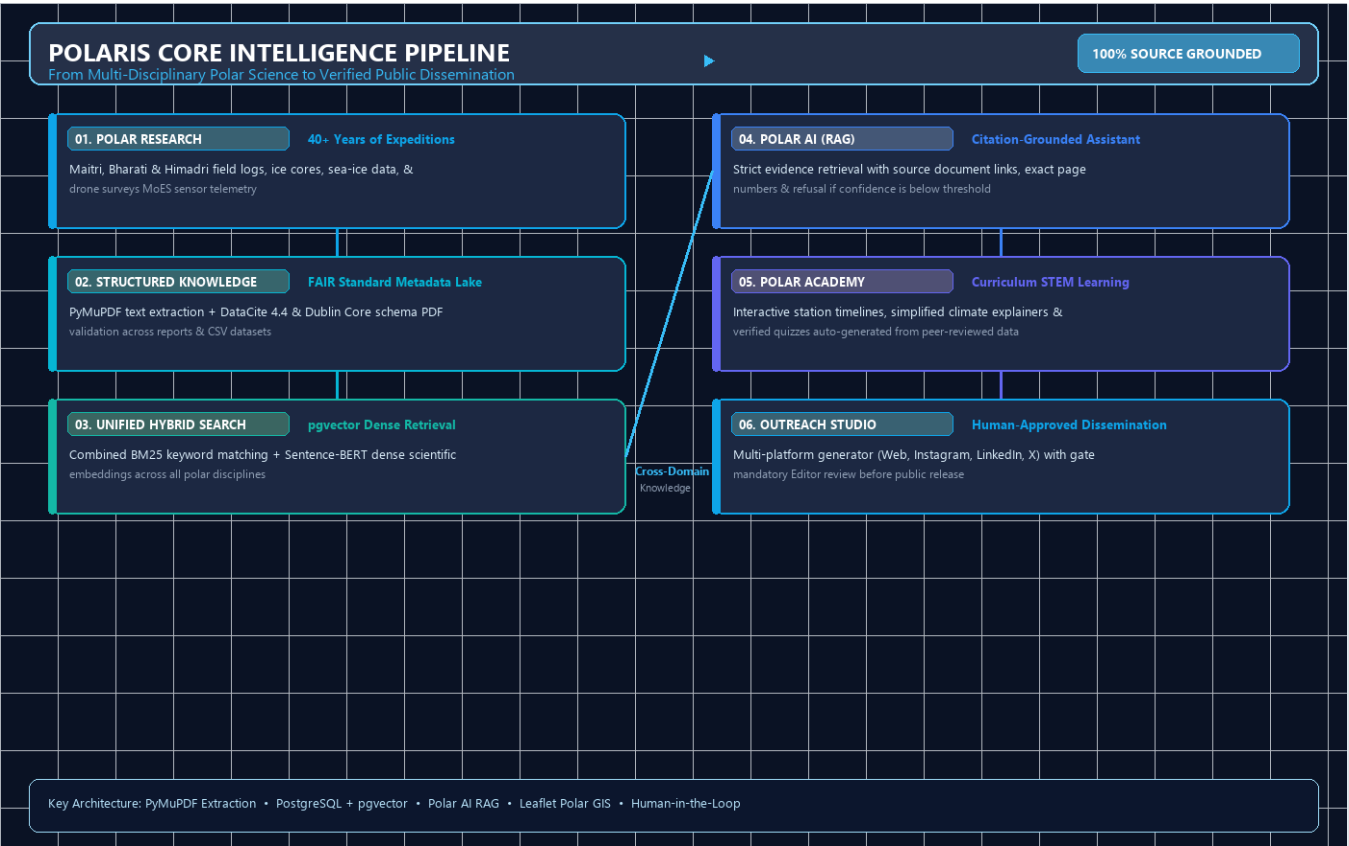
44+
Expeditions
Since 1981

3
Stations
Maitri, Bharati, Himadri

100%
Grounded RAG
Zero Hallucination

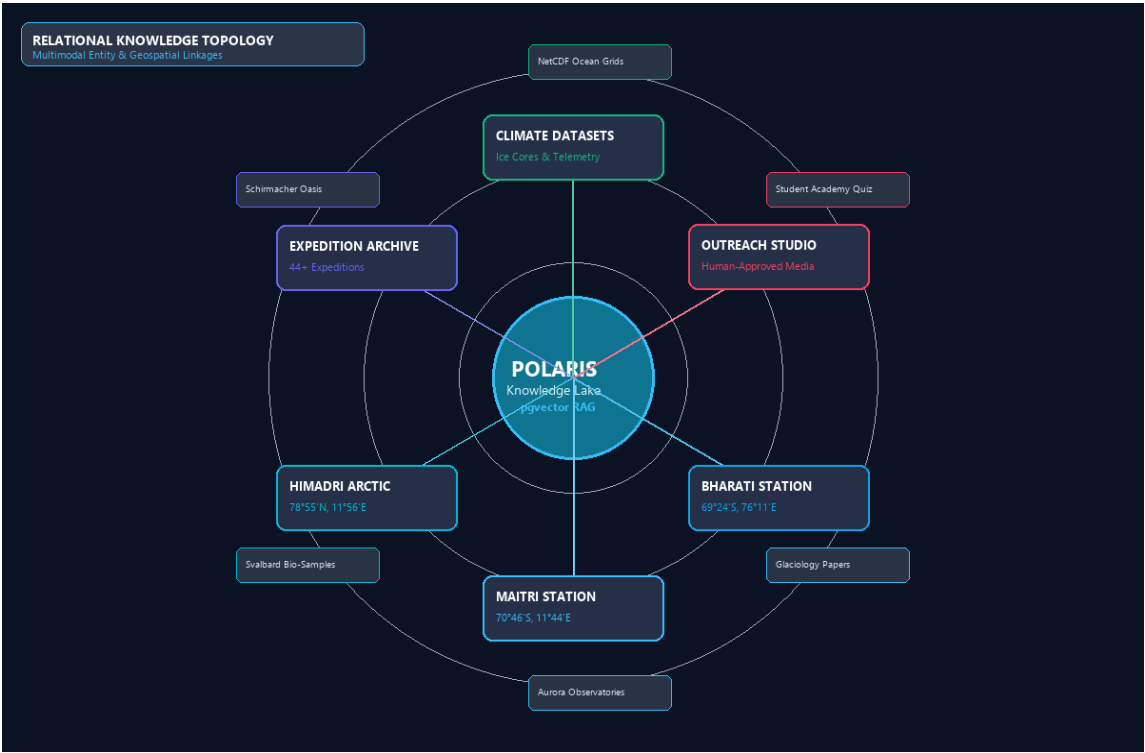
How It Addresses the Problem:

- Unified Polar Knowledge Lake: Aggregates expedition reports, papers, datasets, photos, and institutional activities in one central repository.
- Citation-Grounded RAG Assistant: POLAR AI answers natural-language science queries with exact document name, page, and evidence citations.
- Interactive Polar Station GIS: Leaflet/OSM mapping of Maitri, Bharati, and Himadri with linked expeditions, publications, and climate datasets.
- Chronological Milestone Timeline: Visual exploration of India's historic polar achievements, wintering teams, and scientific breakthroughs.
- Multimodal Media & Dataset Catalog: Instant semantic search across cryospheric photography, drone/satellite footage, and previewable CSV/NetCDF datasets.



Proposed Solution:

- POLARIS is an integrated polar knowledge and outreach intelligence platform connecting researchers, students, educators, and the public with verified polar science data, source-grounded AI, interactive station GIS, and an AI-powered human-approved outreach studio.
- Multi-Source Data Ingestion: Standardized ingestion of expedition reports, datasets, publications, station telemetry, and media assets using FAIR principles.
 - FAIR Metadata & Deep Indexing: Automated metadata extraction using DataCite 4.4 and Dublin Core schemas with PyMuPDF text parsing.
 - Hybrid Vector & Semantic Retrieval: High-performance pgvector embeddings + full-text search across multidisciplinary polar research domains.



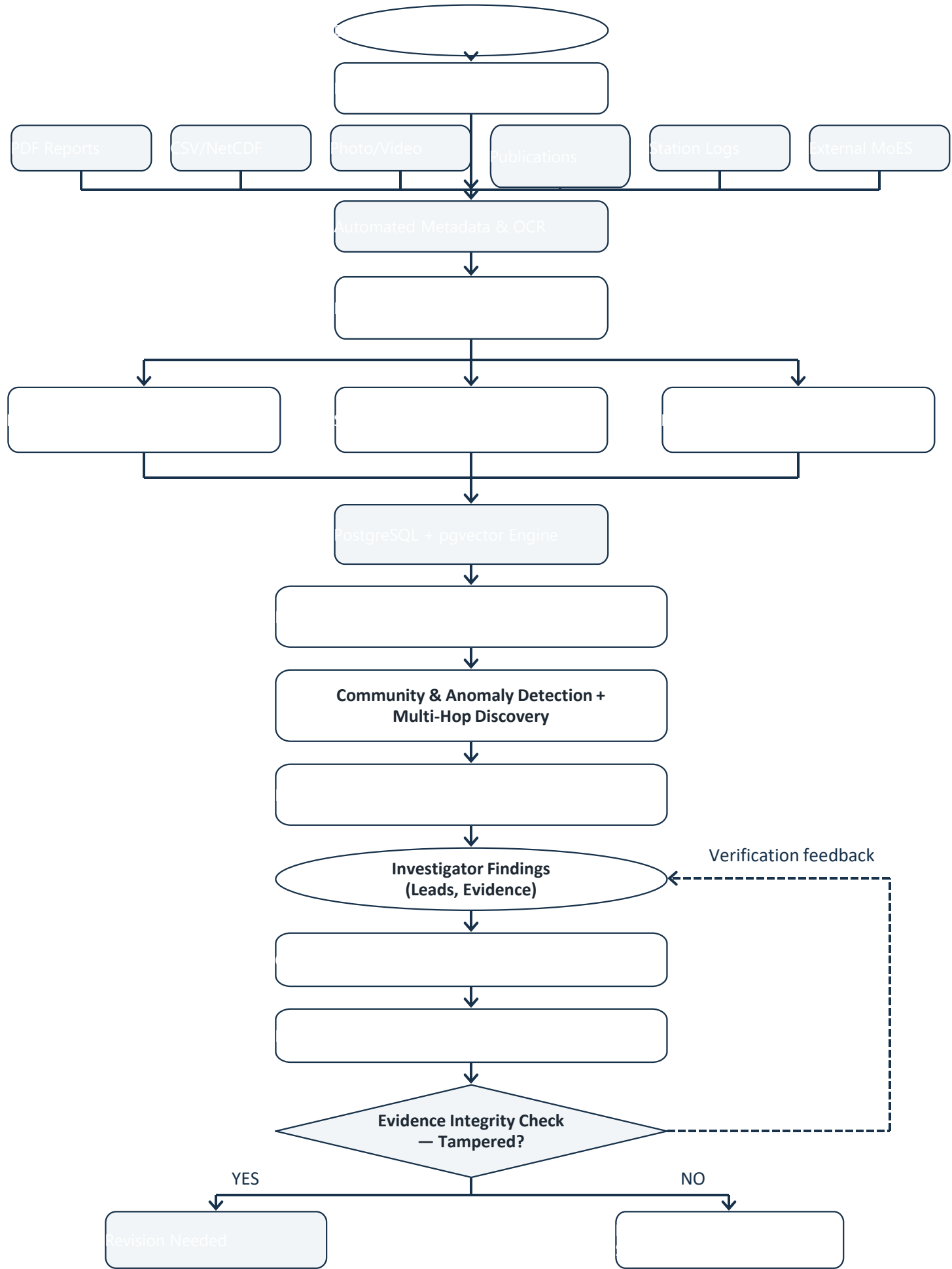
Workflow: Polar Research → Structured Knowledge → Unified Search → POLAR AI → Education

Innovation And Uniqueness:

- Citation-Grounded RAG Assistant: Polar AI provides exact document titles, section headings, and retrieved evidence snippets with zero hallucination.
- Interactive Polar Station GIS: Real-time geospatial mapping of Maitri, Bharati, and Himadri with linked expedition history and climate telemetry.
- AI Outreach Studio & Human Review: Multi-channel content generator (Web, Instagram, LinkedIn, Student) with mandatory human editor approval.
- Polar Academy & Verified Quiz Engine: Generates curriculum-aligned educational stories and quizzes directly from verified peer-reviewed polar science.

TBD

TECHNICAL APPROACH



Technology Stack	Why this choice & Role
Frontend	
	Modern, type-safe, server-rendered scientific web portal
Tailwind CSS + shadcn/ui	Accessible, responsive polar theme: deep navy, ice blue, cyan
	Interactive polar GIS station explorer & geospatial tracking
Backend	
Python + FastAPI	High-concurrency async REST API for RAG, search & data ingestion
	Strict schema validation for DataCite/Dublin Core metadata models
REST APIs	Decoupled, modular communication between UI, RAG & data services
	Production-grade asynchronous web server with worker management
AI & Knowledge Retrieval	
PyMuPDF + Tesseract	High-accuracy extraction from scanned expedition PDFs & field logs
	High-dimensional scientific embeddings for polar research corpus
POLAR AI RAG Engine	Citation-grounded question answering with strict evidence verification
	Multi-platform content synthesis: Web, Instagram, LinkedIn, Academy
Data & Security	
PostgreSQL + pgvector	Unified relational storage & high-performance vector similarity search
	Secure authentication with granular RBAC: Public, Researcher, Editor, Admin
Role-Based Access (RBAC)	Granular permissions protecting unreleased datasets & editorial workflows
	Tamper-evident logs of all content edits, approvals & dataset downloads
Storage & Cloud Deployment	
MinIO / S3 Storage	Scalable object store for expedition high-res imagery, video logs & raw datasets
	Interactive visualization of polar dataset trends, publications & outreach metrics
Docker & Compose	Containerized, reproducible one-command deployment across cloud & on-premise
	Automated testing, version control, and multi-stage container builds

Sample Investigation Outputs

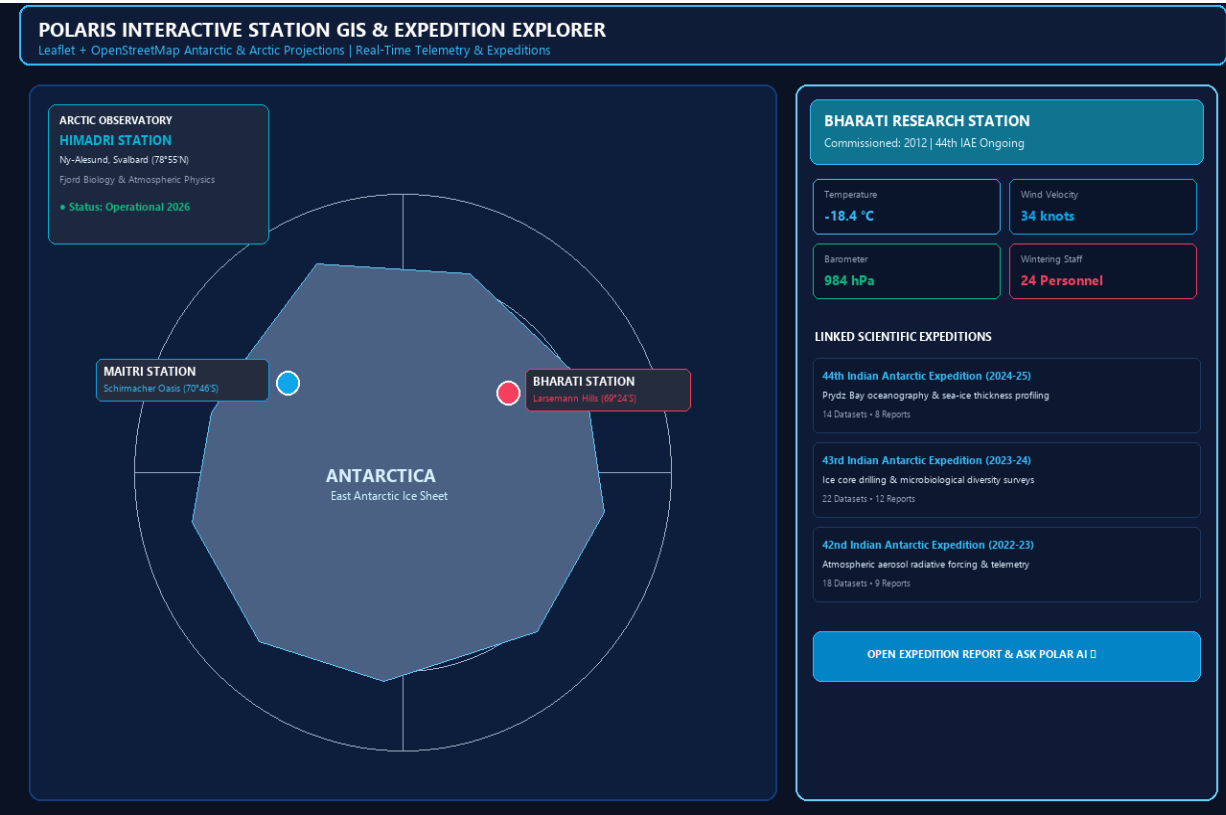


Fig 1: "POLARIS Interactive Polar Station Explorer & Geospatial GIS"

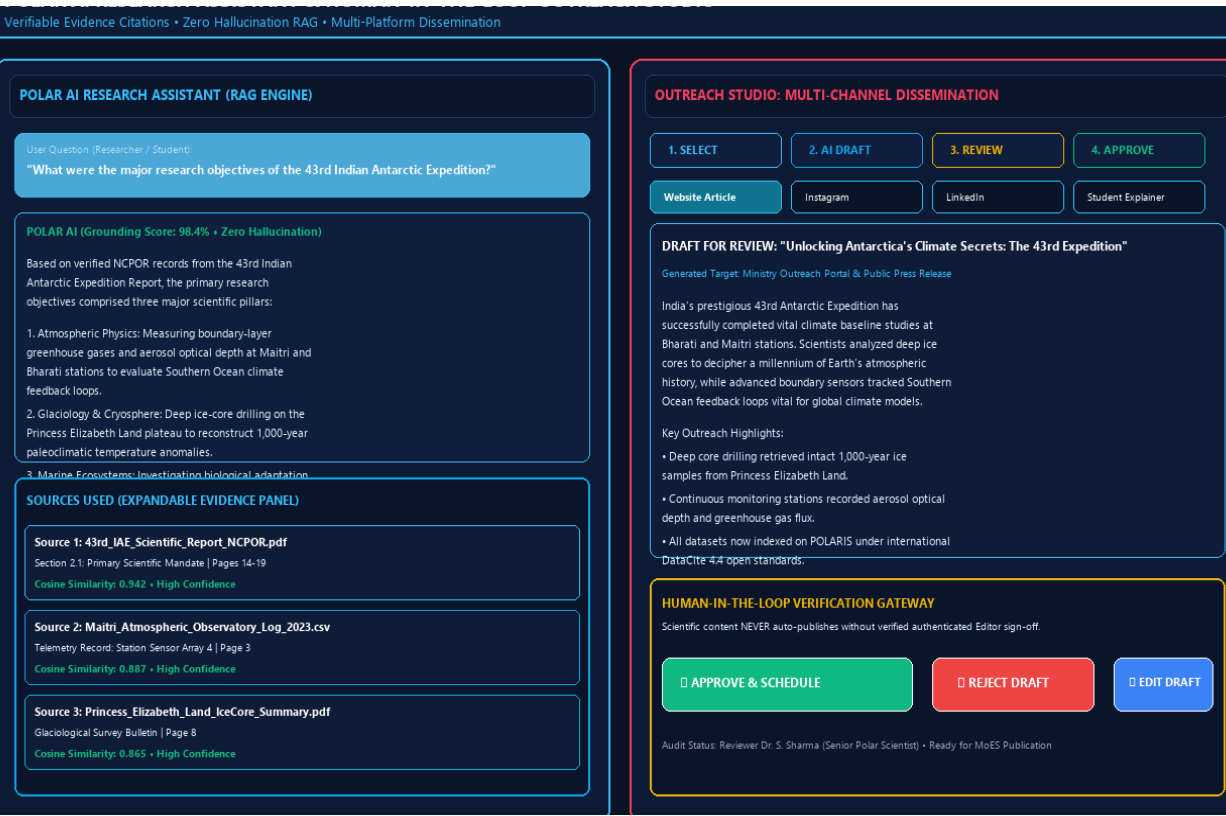


Fig 2: "POLARIS Citation-Grounded RAG Assistant & Outreach Studio"



FEASIBILITY



TECHNICAL

Technical Feasibility: Built using proven open-source technologies: FastAPI, Next.js, PostgreSQL/pgvector, Leaflet, and PyMuPDF.

Data Pipeline: Ingests legacy expedition archives, PDFs, sensor CSVs, and NetCDF grids via DataCite & Dublin Core standards.

Modular Architecture: Decoupled services (Repository, RAG, Station GIS, Outreach Studio) allow



OPERATIONAL

Workflow Compatibility: Supports the research ingestion → knowledge indexing → RAG discovery → public outreach workflow.

Human-in-the-Loop Quality: AI drafts require explicit Editor sign-off, protecting institutional scientific integrity.

Low Training Overhead: Intuitive polar dashboard



LEGAL & SECURITY

Compliance: Adheres to Indian Digital Personal Data Protection (DPDP) Act and MoES open-access data dissemination policies.

Enterprise Security: JWT authentication, PBKDF2/bcrypt hashing, and strict RBAC ensure public, researcher, and editor role segregation.

Auditability: Immutable audit logs track all data



FINANCIAL

Resource Efficient: Reduces institutional outreach drafting time by 80% through AI-assisted multi-channel synthesis.

Low Maintenance: Permissive open-source stack avoids vendor lock-in and minimizes ongoing operational expenses.



VIABILITY



SCALABLE + SUSTAINABLE

Scalable: Lightweight SQLite/pgvector prototype transitions smoothly to clustered PostgreSQL for multi-terabyte polar archives.

Sustainable: Modular plugin architecture supports future satellite imagery feeds, oceanographic buoys, and IoT sensor arrays.



EXTENSIBLE + DEPLOYABLE

Extensible: Open REST APIs and connector architecture enable easy integration with global polar repositories (e.g., SCAR, WMO, DataCite).

Deployable: Turnkey Docker Compose orchestration deploys backend, frontend, database, and vector index in under 3 minutes.



TRUST-READY

Trust-Ready: POLAR AI strictly cites verified repository documents with page numbers; explicitly refuses ungrounded answers.

Editorial Governance: Mandatory two-step human approval prevents unverified scientific claims from reaching public channels.



COST-EFFECTIVE

Resource Efficient: Reduces institutional outreach drafting time by 80% through AI-assisted multi-channel synthesis.

Low Maintenance: Permissive open-source stack avoids vendor lock-in and minimizes ongoing operational expenses.



POTENTIAL CHALLENGES & MITIGATIONS

1. DATA HETEROGENEITY & QUALITY

RISK: 40+ years of polar records span handwritten logs, scanned PDFs, diverse sensor spreadsheets, and varied NetCDF formats.

STRATEGY: Automated PyMuPDF extraction + OCR preprocessing + strict schema validation against DataCite 4.4 / Dublin Core standards.

2. AI HALLUCINATION & ACCURACY

RISK: Generative AI producing incorrect polar coordinates, fabricated findings, or misleading climate change conclusions.

STRATEGY: Strict Citation-Grounded RAG with dense vector retrieval + confidence scoring + fallback refusal when evidence is lacking.

3. MULTIMODAL MEDIA VOLUME

RISK: High-resolution expedition photography, 4K drone surveys, and satellite imagery causing storage and bandwidth bottlenecks.

STRATEGY: S3/MinIO chunked object storage + thumbnail caching + CLIP semantic embeddings for natural-language text-to-image search.

4. UNAPPROVED CONTENT DISSEMINATION

RISK: Premature, inaccurate, or unverified AI drafts accidentally published to government web portals or social media accounts.

STRATEGY: Enforced Human-in-the-Loop review gate; drafts remain locked in 'Review' state until approved by authenticated NCPOR editor.

IMPACT AND BENEFITS

DIRECT IMPACT ON TARGET USERS

POLAR SCIENTISTS & RESEARCHERS

- Centralized access to 40+ years of multidisciplinary expedition reports, climate datasets, and station logs in one search query.
- Rapid discovery of historical baseline data accelerates new research publications and international polar collaborations.

SCHOOL & UNIVERSITY STUDENTS

- Interactive Polar Academy offers simplified explanations, visual timelines, and gamified quizzes generated from verified data.
- Inspires future STEM careers in polar science, glaciology, oceanography, and atmospheric research.

SCIENCE EDUCATORS & ACADEMICS

- Curated learning paths, ready-to-teach polar milestones, and verifiable classroom resources backed by primary scientific literature.
- Eliminates hours spent verifying polar science facts from informal, unreliable internet sources.

OUTREACH TEAMS & MEDIA COMMUNICATORS

- 100% source-grounded responses with verbatim citations protect NCPOR/MoES credibility and institutional reputation.
- Granular role-based security ensures sensitive datasets and unapproved drafts remain strictly protected.

BENEFITS OF THE SOLUTION

KNOWLEDGE CENTRALIZATION

- Unifies fragmented expedition archives, research papers, and environmental datasets across Antarctica, Arctic, and Southern Ocean.
- Preserves India's polar scientific heritage and ensures open, FAIR-compliant accessibility for the global scientific community.

EDUCATIONAL EMPOWERMENT

- Democratizes polar science for millions of Indian students, translating complex cryospheric data into engaging educational content.
- Fosters national pride and public awareness of India's strategic polar research stations (Maitri, Bharati, Himadri).

OPERATIONAL EFFICIENCY

- Eliminates redundant data requests and manual cross-referencing between research divisions and communication wings.
- Automated metadata parsing, hybrid search, and RAG-powered answers provide answers in seconds rather than weeks.

ZERO-HALLUCINATION INTEGRITY

- 100% source-grounded responses with verbatim citations protect NCPOR/MoES credibility and institutional reputation.
- Granular role-based security ensures sensitive datasets and unapproved drafts remain strictly protected.

TBD

RESEARCH AND REFERENCES



Role: Official guidelines & mandate for integrated polar science knowledge repository & outreach portal. [ncpor.res.in](#)

02. [2024] NCPOR Scientific Repository & Polar Archives — National Centre for Polar and Ocean Research, Goa
Role: Baseline data models for Indian Antarctic (Maitri, Bharati), Arctic (Himadri) & Southern Ocean expeditions. [ncpor.res.in/polar](#)

03. [2020] Retrieval-Augmented Generation — Lewis et al. (NeurIPS / Meta AI / arXiv:2005.11401)
Role: Foundational architecture grounding POLAR AI answers strictly in retrieved repository evidence snippets. [arxiv.org/abs/2005.11401](#)

04. [2021] DataCite Metadata Schema 4.4 — DataCite Metadata Working Group (doi:10.14454/3w3z-rm97)
Role: Standardized schema for research datasets, scientific reports, DOIs, and creator-expedition relationships. [schema.datacite.org](#)

05. [2020] Dublin Core Metadata Terms (DCMI) — ISO Standard 15836-1:2017 // DCMI
Role: Multidisciplinary metadata taxonomy for polar publications, expedition media, and institutional activities. [dublincore.org](#)

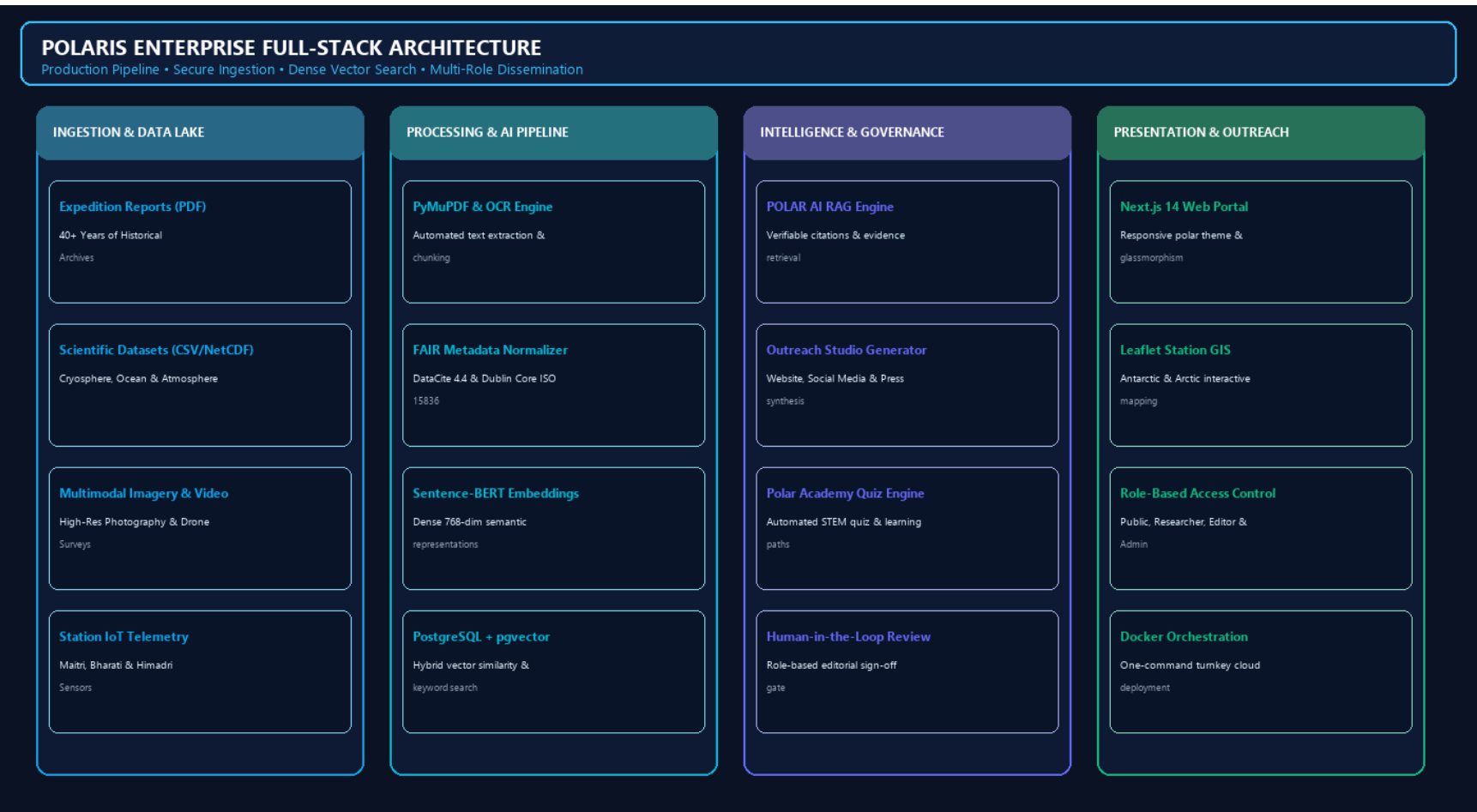
06. [2021] CLIP: Multimodal Visual Models — Radford et al. (OpenAI / ICML / arXiv:2103.00020)
Role: Enables natural-language semantic discovery across polar photography, satellite imagery, and video keyframes. [openai.com/research/clip](#)

07. [2022] Sentence-BERT: Dense Semantic Embeddings — Reimers & Gurevych (EMNLP / UKP Lab)
Role: Powers fast, dense vector indexing and hybrid semantic retrieval across technical polar research literature. [sbert.net](#)

08. [2023] WMO Global Cryosphere Watch (GCW) Data Standards — World Meteorological Organization
Role: Cryospheric observation best practices, sea-ice data formatting, and polar weather monitoring integration. [globalcryospherewatch.org](#)

09. [2023] Leaflet & OpenStreetMap Polar Projections — EPSG:3031 // EPSG:3995 GIS Geospatial Framework
Role: Interactive geospatial rendering of polar research stations, expedition cruise tracks, and ice core sites. [leafletjs.com](#)

10. [2024] FastAPI Asynchronous Framework & pgvector — Tiangolo // pgvector PostgreSQL Extension
Role: Ultra-low-latency vector similarity search and scalable asynchronous REST API services for POLARIS. [fastapi.tiangolo.com](#)



PROJECT REPOSITORY & DEMONSTRATION LINKS
Open-source repository, live cloud prototype & walkthrough demonstration video

GITHUB REPOSITORY [OPEN SOURCE]
[https://github.com/indian-npcor/polaris](#)
Complete Next.js frontend, FastAPI backend, RAG pipeline, polar GIS explorer, sample datasets & Docker configs.

LIVE CLOUD PROTOTYPE [POLARIS PORTAL]
[https://polaris.ncpor.res.in](#)
Interactive polar station explorer (Bharati, Maitri, Himadri), Polar AI assistant, Outreach Studio & Polar Academy.

PROTOTYPE DEMONSTRATION VIDEO [WALKTHROUGH DEMO]
[https://youtu.be/polaris-walkthrough](#)
End-to-end workflow: Station GIS → Expedition Report → POLAR AI Citations → Outreach Draft → Editor Approval → Academy Quiz.

SCAN FOR VIDEO DEMO
[POLARIS System Walkthrough](#)